Ogbonna Arinze

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SUMMARY

Senior Fullstack Engineer with 8 years of professional experience building AI-powered B2B SaaS products across healthcare, fintech, civic tech, and energy sectors. Deep expertise in React, TypeScript, and TailwindCSS on the frontend, with production backend experience in NestJS, NodeJS, PostgreSQL, and Docker. Proven track record integrating LLM and AI models (OpenAI, LangChain) into user-facing products, including production RAG systems, multi-agent pipelines, and LLM observability with Langfuse. Product-minded engineer who thrives in fast-paced startup environments, converting ideas into performant, reliable, and well-tested code.

CORE COMPETENCIES

- **AI & LLM Integration:** RAG pipelines, OpenAI API, LangChain, Vercel AI SDK, multi-agent systems (Mastra), vector search (pgvector, Chroma), embeddings, Langfuse observability, and agentic workflows.
- **Frontend:** React, Next.js, TypeScript, Tailwind CSS, Vue.js, Redux, Zustand, Radix UI, SSR/SSG, and design systems.
- **Backend & Data:** NestJS, NodeJS REST APIs, PostgreSQL, MongoDB, Prisma, GraphQL, JWT authentication, RBAC, and SSE streaming.
- **Infrastructure & DevOps:** Docker, Docker Compose, GitHub Actions, CI/CD pipelines, Nx monorepos, AWS, and Vercel.
- **Testing & Quality:** Playwright, Jest, Cypress, Allure Reports, Storybook, ESLint, and CI-driven QA.
- **Product & Leadership:** Full product ownership, technical architecture reviews, mentorship, Agile/Scrum, and product-minded development.

WORK EXPERIENCE

Genentech (Roche) — San Francisco, CA (Remote)
Frontend Engineer — *May 2023 - Present*

- Co-developed a full-stack **Next.js chat widget** powered by a **RAG system** using **OpenAI GPT models**, **LangChain**, **Vercel AI SDK**, and **Chroma** vector database, enabling users to connect data sources and query them in real time via a conversational interface.
- Spearheaded the redesign of the main clinical trial data product, migrating from a **Vue.js** web application to a **React** Chrome extension, improving workflows for clinical site integrations and user adoption.
- Architected a **reusable design system** in **React** and **Radix UI** with **Tailwind CSS**, improving UI consistency, accessibility (WCAG), and developer velocity across multiple product teams.
- Implemented **JWT-based authentication** and **role-based access control** in **Next.js** with **TypeScript** and **React Query**, using HTTP-only cookies for secure token handling and permission-based UI rendering.
- Introduced **Redux** for state management and **React Query** for request caching, refactoring legacy class components to functional components and streamlining data flow across the application.
- Optimized application performance by implementing **lazy loading, code splitting**, and **CDN asset delivery**, reducing load times significantly as measured via **Lighthouse** and Chrome DevTools.
- Increased release reliability by integrating **Playwright end-to-end tests** into the **CI/CD pipeline** with **GitHub Actions**, automating reporting with **Allure Reports** and configuring Slack notifications for build visibility.
- Mentored a team of two junior developers (interns), fostering a culture of technical excellence and code quality.

Descasio — Lagos, Nigeria (Hybrid)
Senior Frontend Developer — *May 2021 - Mar 2023*

- Led the design and implementation of an intuitive **React** user interface for a **B2B SaaS product**, achieving a **25% boost in user engagement** and collaborating with architects to create a scalable frontend architecture that accelerated development by **30%**.
- Optimized frontend performance by profiling render bottlenecks with **React Profiler** and **Lighthouse**, reducing bundle size by **30-40%** through rollup-plugin-visualizer and webpack-bundle-analyzer, achieving a **35% faster load time**.
- Drove a **40% improvement** in application performance by optimizing caching mechanisms across the React application.
- Enhanced development workflow by introducing **Jest** and **Enzyme** testing frameworks, leading to a **20% decrease** in bug reports.
- Established engineering standards using **ESLint, Prettier, Husky, Lint-staged**, and **GitHub Actions**, reducing errors and accelerating CI/CD processes.
- Mentored junior engineers on frontend best practices, fostering high-quality, maintainable code across projects.

ActivEdge Technologies — Lagos, Nigeria
Senior React Developer — *Oct 2019 - Sep 2021*

- Built a high-volume **bank Settlement Solution** in a team of 3, handling thousands to millions of daily transactions and boosting settlement rates by **80%**.
- Developed critical payment modules (MasterCard, VISA, Express-cash, E-commerce) using **React**, **Redux**, **RxJS** for reactive data streaming, and **Axios** for API integration.
- Adopted the native **EventSource API** for real-time data streaming, replacing socket.io for faster updates and a more responsive interface.
- Built dynamic data visualization dashboards using **React** and **react-chartjs** with **Redux Toolkit** and **Ant Design**.

Founders Lair Beijing, China (Remote)
Full Stack Engineer (Contract) *Oct 2020 - Feb 2021*

- Developed **NodeJS RESTful APIs** and translated business requirements into efficient **MongoDB** database models and schemas.
- Transformed Figma designs into reusable **React** components and engineered a **CI/CD pipeline** for automated deployment to **Heroku** and **AWS**.
- Mentored two interns through frontend and backend projects, providing technical guidance and task management.

SureTrade Business Solutions Nigeria
Frontend Developer *Mar 2018 - Oct 2019*

- Designed **MongoDB** database schemas and implemented server-side logic in **NodeJS**, building and testing **REST API** endpoints for multiple client-facing applications.
- Developed **SureEdu**, a school management portal in a team of 5, handling staff payroll and attendance tracking, student records (exams, reports, attendance), and guardian communications.
- Responsible for full-stack integration, connecting the **JavaScript** frontend to **NodeJS** backend services, developing user use cases, and converting business requirements into database schemas and models.

PROJECTS

OurNigeria ournigeria.ng
Senior Engineer *Feb 2026 - Present*

AI-powered civic platform for Nigerian budget analysis and corruption tracking. 700+ budget documents, 708K+ vector embeddings, 37 states.

- Architected the entire platform as an **Nx monorepo** managing **6 applications**: **NestJS** API backend, **Next.js** citizen-facing web app with **React 19** and **Tailwind CSS v4**, Next.js admin dashboard, NestJS document ingestion pipeline, Next.js marketing landing page, and a Remotion video generation service.
- Designed and implemented a **multi-agent AI system** using the **Mastra** framework with 5 specialized agents and a three-stage intent classification pipeline (keyword scoring, LLM classification with 5-min cache, inferred fallback), powered by a **hybrid RAG pipeline** combining **pgvector** similarity search with BM25 full-text search, Reciprocal Rank Fusion, and **Cohere** reranking over **708K+ vector embeddings**.
- Built the **PostgreSQL 16** database layer with **Prisma v7** and pgvector extension for vector storage, implemented **SSE streaming** for real-time chat responses with 15s keepalive heartbeats, and token-budgeted context management with conversation summarization.
- Configured **CI/CD** with **GitHub Actions** for automated builds, **Playwright** E2E tests across 4 test projects, and **Docker Compose** production deployment with multi-stage Dockerfiles, health checks, and memory limits.
- Integrated **Langfuse** for LLM observability and scoring, **Infisical** for secrets management across environments, and built 22 interactive chart types with **Recharts** auto-generated from AI responses.

EDUCATION

University of Nigeria Nsukka, Enugu
BSc Computer Science - 4.1 CGPA *Graduation Date: Sep 2019*